Chapter 2 · Story

A brighter poster. More visitors. A cause?

An original story at the Common Ground repair café.



The poster that “worked” · c02s01

A fictional story. A real question about people.

What would you do?

Which statement fits Leo’s evidence?

A The poster caused the increase.

B The poster had no effect.

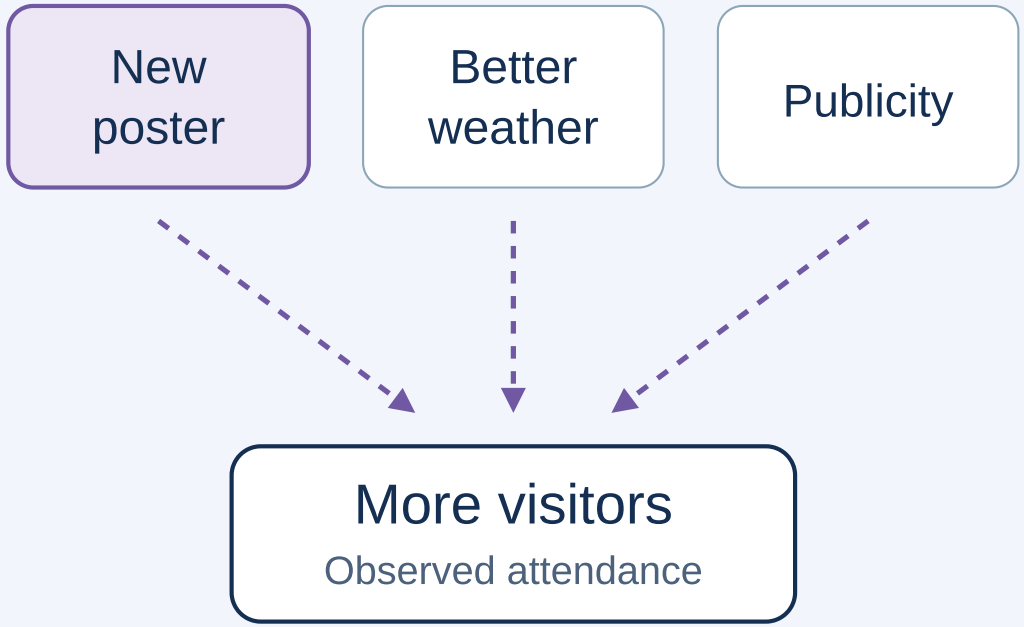
C Attendance rose; several causes remain possible.

Suggested answer: C. Saying that attendance rose preserves the observation. Leaving the cause open reflects the limits of the comparison.

Comparison makes a claim testable.

- **Correlation**: things vary together.
- **Confound**: another factor could explain the pattern.
- **Random assignment**: allocate conditions by chance.
- **Random sampling**: select people from a population by chance.

Several changes, one outcome



Dashed paths: possible explanations

A fair comparison separates explanations.

Several changes can fit the same before-and-after pattern.

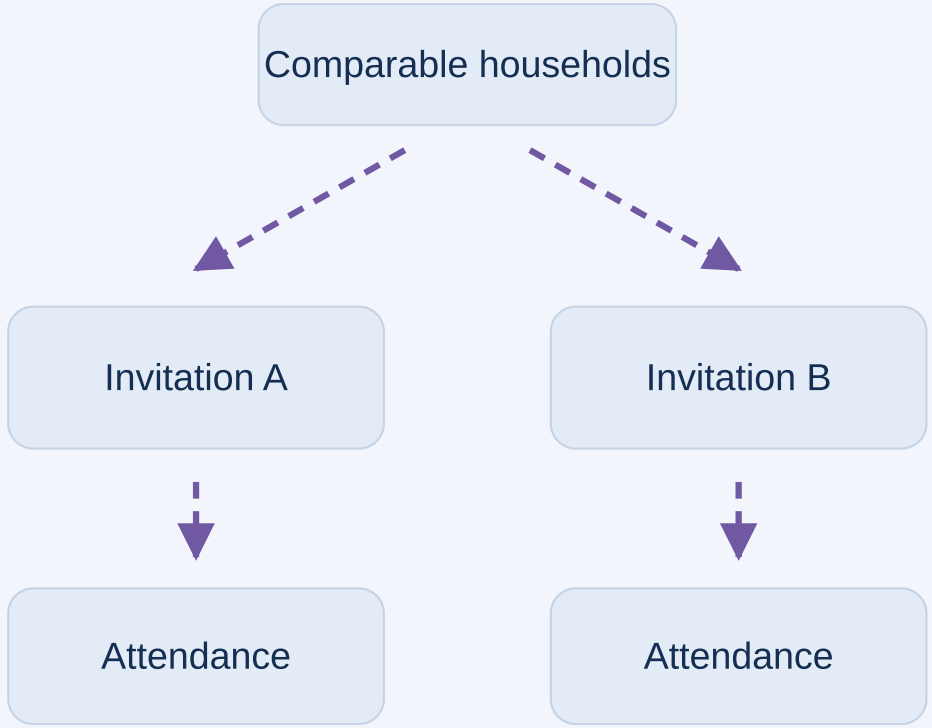
Ask how much—and how uncertain.

Size: 12% versus 15% is a difference of three percentage points.

Uncertainty: another sample may give a different estimate.

Statistical significance: judged against a model, often assuming no group difference.

Significance does not establish practical importance; nonsignificance does not prove zero effect.

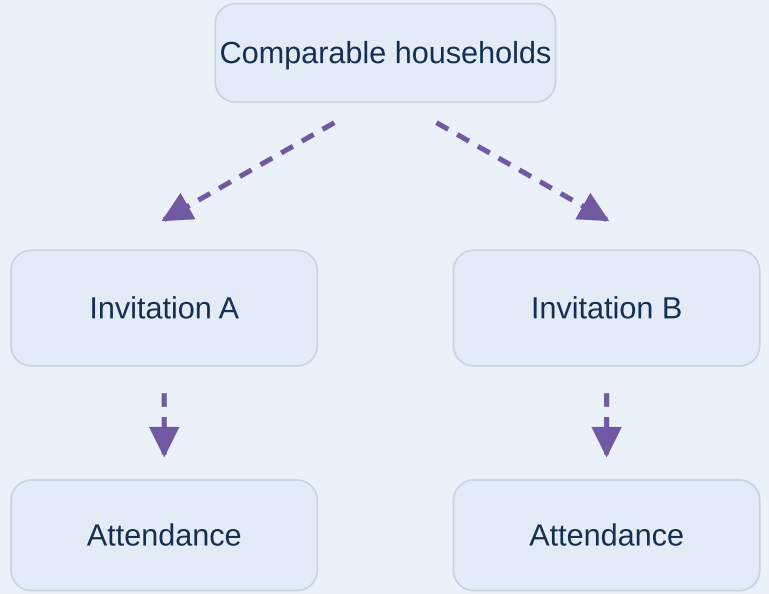


Allocate by chance. Compare like with like.

Random assignment supports a fairer comparison.

What can we really conclude?

Replication repeats a test to examine how dependable a finding is. Independent tests help us learn whether a result holds up elsewhere. Preregistration records the planned methods and analysis before results are known. That advance record reduces some opportunities for selective reporting. Preregistration cannot rescue a poor design. With a sound design, a larger sample can narrow the uncertainty around an estimate. More observations cannot fix a systematically unfair comparison. Sharing the study's methods and materials lets others inspect how it was done. A memorable study is a starting point for evaluation, rather than a final verdict.



Allocate by chance. Compare like with like.

Conceptual illustration; no research data are plotted.

Try the idea somewhere new.

Which invitation comparison is strongest?

A Design A in winter; design B in summer.

B Randomize designs for one event; define attendance in advance.

C Design A for existing fans; design B for newcomers.

Suggested answer: B. Randomizing invitations for one event improves the comparison. Also check whether households share invitations.

Close your notes. Explain it.

> How does random assignment differ from random sampling?

> Why is before-and-after evidence insufficient?

1. Random assignment allocates people to conditions. Random sampling selects people from a population.

2. Other conditions can change over time and produce the same before-and-after pattern.



Maya, Sam and Leo at the repair café.

Carry one idea into your week.

Find an everyday causal claim. Name its outcome, one competing explanation, and a feasible comparison. State whose behavior the result would represent.

Revisit tomorrow and in a week.
Explain it without notes, then use a different example.



Take the question into a different setting.